Readable Depth Without the Calibration Map: Formula Placement on Heavy-Hex Devices, Measured by a Circuit's Own Conserved Charge

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Abstract

Compilers for today's quantum processors choose a layout by reading the day's calibration: error rates per qubit and per coupler. A classical computer offers no such map, and a map is stale as soon as it is read. We ask how deep a circuit stays readable when no placement reads a quality number. We place a ring of CZ gates on IBM's heavy-hex devices in three ways that see only the coupling graph, the basis and which elements have failed: a formula that puts the ring on a lattice cycle (the *rule*), the ring folded onto the same qubits (*zipper*), and Qiskit's level-2 search given the coupling map without error rates (*blind Qiskit*). As the workload we use the kicked Ising ring at $J = \pi/4$, whose exact answer is known at any size, and as the probe a weight-2 charge conserved by the circuit itself: its decay, read with a four-input combination that cancels readout and first-order relaxation, gives the per-step damping without any calibration data. In declared tests on `ibm_marrakesh` and `ibm_kingston` (29 October 2026; $n = 20$ and 36; $L = 24$, 50, 100 steps, up to 3600 CZ per circuit) the rule kept more of the charge than both other placements in all 24 comparisons. Its readable depth, the $1/e$ decay length L^*, was 56–57 steps on marrakesh and 81–90 on kingston, against 15–26 for the others: 30–73 steps deeper. In an earlier test with a 16-layer classifier at $n = 13$–24 the rule beat blind Qiskit at 11 of 12 sizes and tied one. The noise model we declared did not hold: one depolarizing rate did not describe the decay on either device, and the zipper's penalty, 3.5–3.7 times the rule's decay rate on both devices, was larger than its gate count and duration predict (2.2–2.4). We report these failures with the results. [TODO: sizes above 36 (ring extension, 30 Oct.); a declared test of the zipper penalty]

1 Introduction

- Map-aware placement is the default (Qiskit's VF2 post-layout and noise-aware routing read error rates). Its gains are measured against the same map that chose the layout; the selection is optimistic by construction (the optimizer's curse [3]), and the map ages.
- The question here: without any quality number, how deep can a circuit be read, and does the placement change that depth? We call a placement *map-free* if it reads only the coupling graph, the basis and the failed elements. Map-aware placement is a special condition and is always labelled so.
- What is new: (i) a map-free comparison of placements by a calibration-free decay measurement on the circuit itself; (ii) the readable depth L^* and the gain ΔL as the reported

quantity; (iii) declared predictions, with the failed ones reported. What is not new: reading noise from conserved quantities [5, 6], layer fidelity and size-independent layer errors [7], the noisy free-fermion solution [8], and the $J = \pi/4$ kicked Ising circuit on IBM devices [11].

- A picture from underwater acoustics guides the reading (Section 2.5). Sound in the ocean fades with distance at a rate set by its frequency and by the water; low tones in the deep sound channel carry for thousands of kilometres. Here the depth of a circuit plays the distance, the Pauli weight of what is read plays the frequency, the placement plays the water, and a conserved charge plays the sound channel. It is an analogy for the reader, not a physical identity.
- Companion work: the formula schedules and their gate counts [1].

2 Setting

2.1 Workload

The kicked Ising ring, one step $S = (R_X(2h)\, Z)^{\otimes n}\, \mathrm{CZ}_{\mathrm{ring}}$, $J = \pi/4$, $h = 0.6$. It is a free-fermion circuit; its outputs are computed exactly at any n by a recursion, which also gives the outputs under uniform depolarizing noise exactly [8]. [TODO: one paragraph with the recursion, from toqb.kising]

2.2 Placements (no quality number read)

- **Rule** (`fold:rule`): the ring on a cycle of the heavy-hex lattice, built by formula from the lattice's cells [1]; n CZ per layer.
- **Zipper** (`fold:zipcyc`): the ring folded onto the same cycle's qubits; $5n/2 - 4$ CZ per layer.
- **Blind Qiskit** (`qiskit:2:blind:template`): Qiskit level 2, `seed_transpiler`=0, given the working couplers in both directions and the basis only.
- Failed elements (error = 1) are removed for every placement; nothing else of the calibration is used.

2.3 Probe P1: the circuit's own conserved charge

$$Q = \sum_j \big(Z_j Z_{j+1} + b\, Y_j Z_{j+1} + c\, Z_j Y_{j+1}\big), \qquad b = c = -1.28608 \ (h = 0.6),$$

the weight-2 conserved charge with the largest ZZ share (a noiseless criterion; the same coefficients for $n = 6$–40). Inputs $|0\cdots0\rangle$, $|1\cdots1\rangle$ and the two Néel states; three measurement settings; every record carries its own readout reference on the physical qubits it measures. The four-input ratio

$$\rho(L) = \frac{[Q(0\cdots0) + Q(1\cdots1) - Q(\text{Néel}) - Q(\overline{\text{Néel}})]_L}{[\,\cdot\,]_{\mathrm{ref}}}$$

cancels readout within a record and the first-order relaxation bias. Under uniform depolarizing noise $\rho(L) = (1 - 4p/3)^{2L}$.

2.4 Readable depth

L^* is the $1/e$ decay length of a one-parameter fit $\rho(L) = e^{-L/L^*}$ over $L = 24, 50, 100$ (weighted least squares on the ρ scale). $\Delta L_X = L^*_{\text{rule}} - L^*_X$ and $\kappa_X = L^*_{\text{rule}}/L^*_X$, the ratio of per-step decay rates. L^* is a property of a weight-2 quantity; wider outputs decay faster (Section 7).

2.5 An acoustic reading of the decay

Under uniform depolarizing noise of rate p per qubit and step, a Pauli string of weight w is multiplied by $(1 - 4p/3)^w$ at every step. The attenuation per step therefore grows with the weight, as the absorption of sound in sea water grows with frequency [9]. Table 1 lists the correspondence we use.

sound in the ocean	a circuit on a device
distance travelled	depth L (steps)
absorption coefficient; attenuation length	per-step decay rate; L^*
frequency (higher tones fade sooner)	Pauli weight of the observable (wider observables fade sooner)
the water (temperature, salinity)	the placement (the zipper absorbs about 3.6 times as much as the rule)
geometric spreading loss	operator spreading: a local output grows in weight before the noise acts on it
the deep sound (SOFAR) channel	a conserved charge: the circuit does not spread it; only the noise damps it
sonar equation and detection range	signal against shot noise; the readable depth at a shot budget

Table 1: The analogy used in this paper. Where it fails: sea-water absorption grows roughly as the square of the frequency, with relaxation peaks, and is an empirical formula; the weight dependence here is linear in the exponent and follows from the noise model.

In the language of the sonar equation [10], a reading at depth L is detected when the signal exceeds twice its shot-noise standard error. With $\rho(L) = e^{-L/L^*}$ and a per-shot spread σ_1 (so that $\text{SE} = \sigma_1/\sqrt{S}$ at S shots),

$$L_{\text{read}}(S) = L^* \ln \frac{\sqrt{S}}{2\sigma_1}.$$

For probe P1, $\sigma_1 \approx 0.2$–0.26, so $L_{\text{read}}(1000) \approx 4L^*$. That lies beyond the depths we measured and is not claimed; we report L^*, which needs no budget and no threshold. The formula does show why the gain matters: ΔL in L^* becomes about four times as many steps of detection range at 1000 shots, and a placement that absorbs κ times as much reaches κ times fewer steps with the same shots.

2.6 Declarations

Every device test was declared (SHA-256 and CET time) before its calibration was fetched; the declarations, records and raw counts are in TOQB [2]. [TODO: table of declarations and job ids]

3 Classifier test without the map (D6)

TOQB's ring classifier, $L = 16$, $n = 13$–24, 64 inputs $\times$ 250 shots, `ibm_kingston`, 28 October 2026; no arm read a quality number and no simulator was used. Rule against blind Qiskit: 11 wins, 1 tie

(at $n = 23$), no loss. Rule against the zipper on the same qubits (even n): 3 wins, 3 ties, no loss. Blind Qiskit kept no measurable signal at 8 of 12 sizes; its circuits ran 22–108 µs against the rule's 5.6–9.3 µs. For reference, with the map (D5 part A, $n = 20$, a different hour): rule 0.943 ± 0.044, zipper 0.600 ± 0.044, map-aware Qiskit 0.765 ± 0.099 (rule − Qiskit within 2 SE). [TODO: table of D6 from TOQB results/device-d6]

4 The noise wall on two devices

	marrakesh $n = 20$	marrakesh $n = 36$	kingston $n = 20$	kingston $n = 36$
L^* rule	56.5 ± 0.9	56.6 ± 0.7	81.3 ± 1.3	89.7 ± 1.1
L^* zipper	15.1 ± 0.4	15.7 ± 0.3	22.6 ± 0.4	25.6 ± 0.3
L^* blind Qiskit	26.4 ± 0.5	15.7 ± 0.3	22.0 ± 0.4	17.0 ± 0.3
ΔL vs zipper	+41.3	+40.9	+58.7	+64.1
ΔL vs blind Qiskit	+30.1	+40.9	+59.3	+72.7
κ zipper (g, t)	3.73 (2.30, 2.27)	3.60 (2.39, 2.25)	3.60 (2.30, 2.24)	3.50 (2.39, 2.22)
rule wins on ρ	6/6	6/6	6/6	6/6

Table 2: Probe P1 on two devices, 29 October 2026. g, t: the zipper's gate and duration ratios to the rule. 1000 shots per circuit; jobs of 92 and 90 s. [TODO: full ρ tables in an appendix]

- **Benchmark number:** the rule won all 24 charge comparisons; $\Delta L = +30$ to $+73$ steps.
- **Device user's number:** at $n = 36$, $L = 100$ (3600 CZ) the rule's charge was 0.178 ± 0.006 (marrakesh) and 0.314 ± 0.006 (kingston); the zipper's and blind Qiskit's were zero within 2 SE.

5 Declared predictions that failed

- **One depolarizing rate.** Neither device followed $\rho = (1 - 4p/3)^{2L}$ with one p within 2 shot-noise SE: marrakesh against its own p from $L = 24$ (1 of 5 inside), kingston against $p = 0.0051$ from the previous day (1 of 6). The per-segment decay rates scatter by 10–20 % with no common direction, and kingston's rate had improved since the previous day; the declared tolerance (shot noise only, about 1 %) left no room for either.
- **Size independence of the rule.** SAME at all three depths on marrakesh; DIFFERENT at $L = 24$ and 50 on kingston, where the $n = 36$ ring was better than the $n = 20$ ring.
- **The structure bracket.** The zipper's κ was declared to lie between its gate ratio and its duration ratio (2.2–2.4). It was 3.5–3.7 on both devices.
- **Probe P2** ($\langle X_0 \rangle$, 250 shots) was at the noise floor on both devices.

6 The formula fixed after the run

The following changes were made after seeing the results. They are fitted to the same data and are evidence only once they predict a later run; they are declared for the next stage.

1. **Self-reference with a model band.** The rule's rate is read from its own $L = 24$ point; deeper points are predicted with the rate $\pm 20\,\%$ (the observed scatter), plus 2 SE of shot noise.

2. **A constant zipper penalty.** $L^*_{\text{zipper}} = L^*_{\text{rule}}/3.6$, with $\kappa \in [3.3, 3.9]$, in place of the structure bracket.

3. **Map-aware, for reference only (special condition).** $1/L^* = c\times$ the median CZ error of the rule's couplers on the day's calibration.

fixed formula	on its own data	fitted on one device, applied to the other
1. own $L = 24$ point, rate $\pm 20\,\%$	8/8	(no device constant)
1b. $n = 20$ ring's $L = 24$ point to the $n = 36$ ring	6/6	(no device constant)
2. zipper $\kappa = 3.6 \pm 0.3$	4/4	4/4
3. map-aware: $1/L^* = c\times$ median CZ error	fits by construction	1/4 (-27 to $+37\,\%$)

Table 3: The fixed formulas on 29 October 2026 (post-hoc). Formula 3 reads the calibration and failed across devices: the day's error map did not predict the readable depth better than the circuit's own short-depth point.

7 Output probe P2

[TODO: P2 tables; the contrast between a ring-averaged weight-2 charge and a single-site output]

8 Discussion

- **Selection and prediction.** The rule's schedules are fixed by formula; nothing is chosen as the best of several candidates by a calibration- or simulation-based score. For a selection of the best of K candidates by an estimate with error scale σ, the prediction is optimistic by up to $\sigma\sqrt{2\ln K}$ [3]; the analogy with the Occam bound [4] is structural. This concerns the reliability of a prediction, not which placement keeps more signal, which we measured separately.

- **Why the duration matters.** Blind Qiskit placed the bare ring at equal gate count, yet its charge decayed 2–5 times faster per step than the rule's, with 5–9 times the duration. [TODO: idle decoherence, scheduling]

- **What a map buys.** D5 part A suggests about 0.3 in the classifier's slope at $n = 20$; between jobs, not declared. [TODO: a declared non-inferiority test with a stale map]

9 Limitations

- One day per device, one ring per size, two sizes. [TODO: sizes above 36]

- The charge is a weight-2 quantity; L^* is not the depth at which an arbitrary output stays readable.

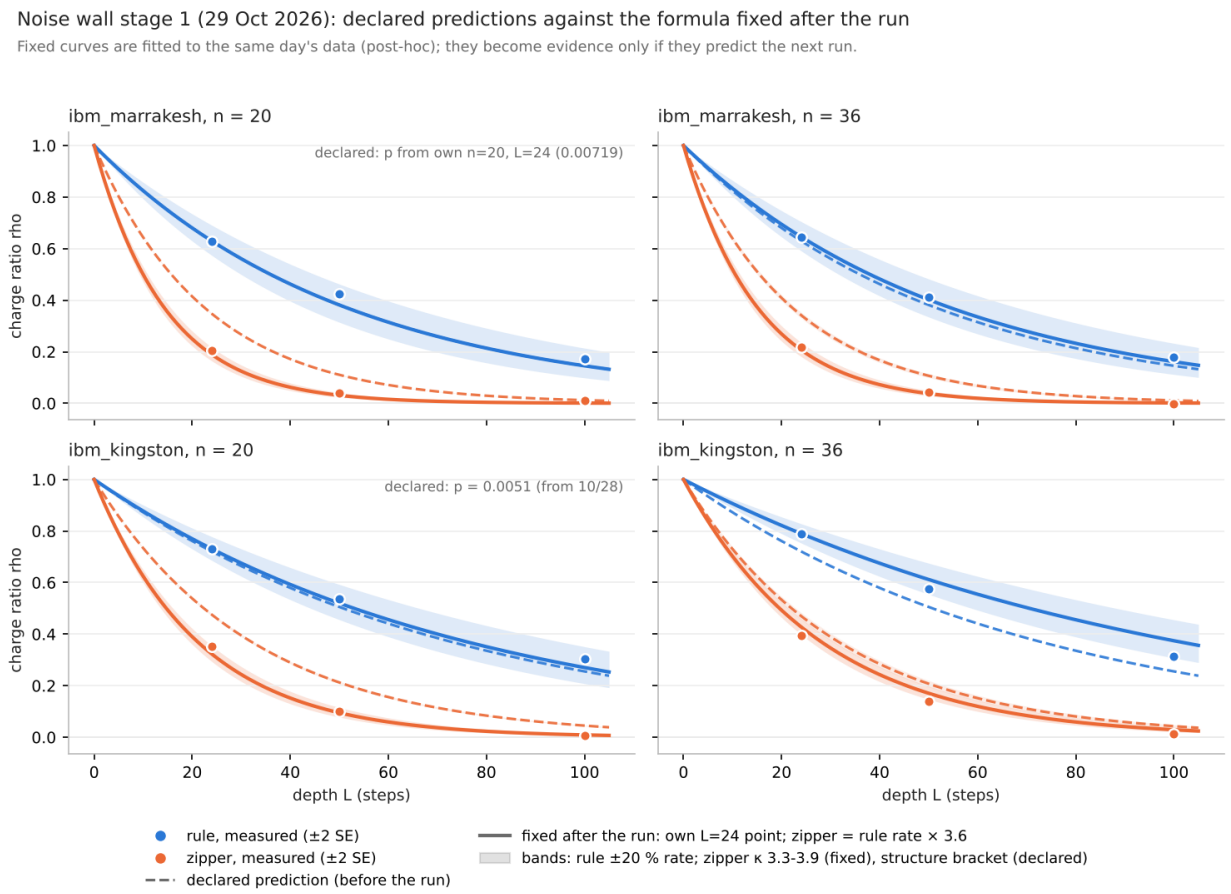


Figure 1: Declared predictions (dashed) against the formula fixed after the run (solid, with bands), for the rule (blue) and the zipper (orange) on both devices and both sizes. Points: measured ρ, ± 2 SE. The largest miss of the declared predictions is the zipper's: its structure ratio (about 2.3) underestimated its penalty (about 3.6) on all four panels. Fitted to the same day's data; script `readable_depth_figures.py`.

- The four-input ratio leaves a second-order relaxation bias.
- Every reference here is classically computable; no classical-hardness claim is made.

10 Reproducibility

TOQB [2]: `results/device-d6-2026-10-28`, `results/noise-wall-1-marrakesh-2026-10-29`, `results/noise-wall-1-kingston-2026-10-29`; declarations in `prereg/`. Circuit files over GitHub's size limit are kept off the repository with their SHA-256.

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device runs used the IBM Quantum Open Plan, including its one-time promotion of 180 minutes of runtime.

References

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